

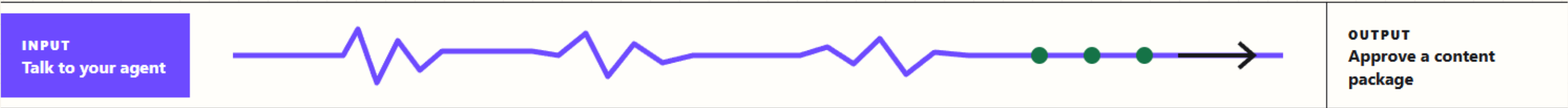
Conversational Systems Atlas

A voice-first SaaS that turns a creator’s spoken intent and trusted market evidence into approved YouTube content packages.

Product
YT Growth Stack

Visual premise
Voice enters left. Approved creative work leaves right.

Architecture stance
Modular monolith first; extract workers when usage proves the need.

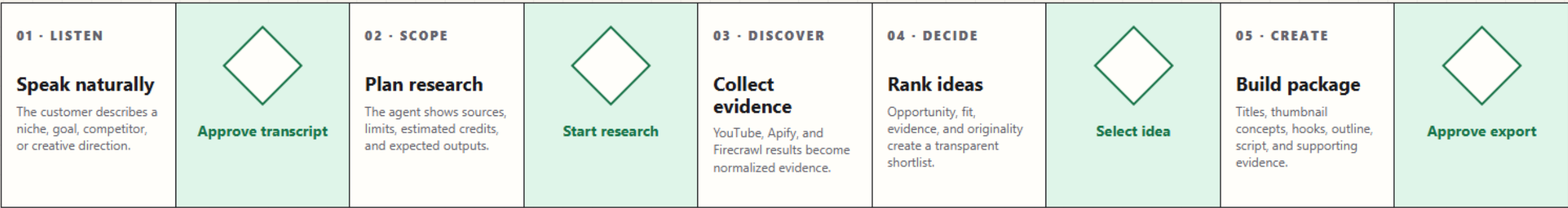


LAYER 01	LAYER 02	LAYER 03	LAYER 04	LAYER 05	LAYER 06	LAYER 07
Experience The customer-facing command centre.	Voice Low-latency conversation with a visible transcript.	Trust Identity, tenancy, roles, and budgets.	Orchestration Durable work outside normal page requests.	Research Owned, competitor, and wider web evidence.	Intelligence Evidence becomes ranked creative direction.	Data + Ops Durable state, files, events, and support.
Next.js + React	WebRTC session	Supabase Auth	Job state machine	YouTube APIs	Pattern extraction	Supabase Postgres
shadcn/ui	GPT-Live	Workspaces + roles	Retries + timeout	Apify Actors	Idea generation	Storage + Realtime
TweakCN tokens	GPT-Realtime-Whisper	Postgres RLS	Cancellation	Firecrawl	Opportunity scoring	Audit logs
Chat + artifact editor	Interrupt + replay	OAuth secrets	Idempotency	Normalization	Package generation	Cost + latency
Realtime progress	Text fallback	Usage limits	Approval events	Source provenance	Quality validation	Admin tools

Security rail Short-lived browser credentials · encrypted server secrets · workspace RLS	Evidence rail Source URL · collected at · method · confidence · immutable artifact version	Human rail Transcript · research scope · idea · package · publishing approval
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From spoken intent to approved content

The agent may research and propose, but the customer controls costly generation and consequential action. Every approval attaches to an immutable version.



Voice is the default, not the only path

Text composer, editable transcripts, keyboard controls, captions, and visible state keep the system accessible and precise.

Approval means a specific version

Any later edit produces a new artifact version and requires reapproval. The audit record identifies who approved what and when.

No autonomous publishing in MVP

The first version ends at export or copy. A later publishing integration requires a separate explicit confirmation and scoped OAuth permission.

The conversation is the control plane

The dashboard keeps speech, evidence, editable artifacts, and approvals visible at the same time.

ConversationLive

Find an underserved angle for a beginner video about AI automation.

I can compare your channel with five selected competitors and supporting web research. Estimated use: 14 credits.

Listening · click or press Space to stop

Working artifactPackage v3

Opportunity

"The first automation a non-technical owner should build" 86 / 100 · strong evidence

Title direction

Stop Learning AI Tools. Build This One Automation First.

Thumbnail concept

One glowing workflow against a wall of crossed-out app icons.

Hook

"Most beginners collect AI tools. The useful ones connect one painful task from start to finish."

Outline + script

Editable sections with inline source markers and regeneration controls.

Ready for customer review

Evidence7 sources

Owned channel

Relevant audience retention and topic history from authorised YouTube data.

Competitor pattern

Repeated beginner framing with high views but low implementation depth.

Web signal

Supporting demand language from articles and product pages.

Why this score?

Strong audience fit, clear evidence, moderate competition, differentiated promise.

OPERATING MODEL

Two loops. Different jobs.

The analysis loop serves customers. Loop engineering serves the repository. Neither removes human judgment.

RUNTIME

Product intelligence loop

Runs when a customer asks the SaaS to create work.

Listen

Research

Generate

Validate

Approve

REPOSITORY

Loop engineering

Runs when coding agents change the product.

Issue + plan

Branch

Implement

Checks + review

Human merge

Hard boundary: agents do not autonomously publish content, expose credentials, alter billing, destroy production data, or merge their own pull requests.

CONCERN	CHOSEN ROLE	WHAT IT DOES NOT DO	CONTROL THAT MATTERS
Next.js	Experience, server endpoints, and initial orchestration shell	Does not hold long HTTP requests open for scraping	Server-only provider clients and explicit module boundaries
Supabase	Auth, Postgres, RLS, Storage, Realtime	Does not automatically create safe tenant policies	RLS tests for every customer-owned table
OpenAI voice	Realtime spoken agent and live transcript	Does not receive permanent API keys in the browser	Short-lived session, configurable models, usage meter
YouTube APIs	Authoritative data for the connected owned channel	Does not provide unlimited quota or every public transcript	Incremental OAuth, quota monitoring, compliance planning
Apify	Permitted public competitor collection	Does not become the source of truth for owned analytics	Async runs, webhook verification, normalized schemas
Firecrawl	Supporting web research beyond YouTube	Does not duplicate the YouTube collector by default	Domain allow/deny rules, provenance, retention
AI pipeline	Patterns, ideas, scoring, packages, validation	Does not approve its own consequential output	Prompt/model versions, evidence links, evals, approvals
Loop engineering	Agent-readable development and PR workflow	Is not the production backend	AGENTS.md, skills, checks, iteration caps, human merge

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YouTube OAuth + quota · developers.google.com/youtube/v3/guides
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LOGICAL ARCHITECTURE IS FINAL · DATABASE SCHEMA AND REPOSITORY BLUEPRINT ARE THE NEXT DESIGN LAYER